

Grounding quality measurement for orchestrated AI reasoning chains: evidence-binding rate as an evaluation primitive for world model systems

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Abstract

AI systems that reason about physical or scientific domains routinely produce outputs containing scientific claims without citing their generating evidence. R_{uncited} — the fraction of scientific sentences in a system’s output that carry no bound citation — is a formal evaluation primitive for AI reasoning quality. Using a reproducible 8-scenario parallel benchmark against the Gemini Interactions API with Google Maps grounding on marine science queries, Maps-only geospatial grounding achieves 60–100% uncited across query types. Injecting a structured orchestrated skill-dispatch step-log as RAG context reduces R_{uncited} to 0%, demonstrating that planning traces eliminate uncited claims by converting open-domain science queries into closed artifact-reference queries. The RAG lift hierarchy (step-log > structured data > narrow context > unstructured) functions as a diagnostic for grounding architecture quality: context that binds the query to a complete reasoning trace eliminates uncited claims; narrowly structured context does not. The extension of this benchmark to AI planning systems whose intermediate representations are required to be grounded in their generating sensor observations is examined.

1 The evidence-binding gap in AI reasoning systems

AI systems that generate natural language outputs about scientific or physical domains make claims that require empirical grounding. When a language model responds to a question about orca encounter likelihood at Haro Strait, it may correctly name the relevant salmon corridor, acoustic monitoring network, and behavioral studies — but without binding any of these claims to their generating observations, prior analyses, or data artifacts. The response is fluent and plausible; it is not evidentially accountable.

This is the evidence-binding gap: the structural disconnection between a system’s stated claims and the observations, methods, and experiments that would make those claims verifiable. The gap is not a matter of factual accuracy per se — a claim may be broadly correct while still carrying no bound evidence. It is a matter of verifiability: a judge, steward, or downstream system cannot determine whether the claim was produced by accessing ground-truth data or by reproducing a plausible-sounding pattern from training.

Prior work on RAG hallucination reduction [4, 5, 15, 10] establishes that retrieval-augmented generation reduces factual error rates and that citation generation can be automated. However, this literature focuses on whether RAG improves accuracy, not on whether the improvement comes from evidence binding — from the system’s outputs being traceable to their generating context — versus improved parametric knowledge activation.

The geospatial grounding baseline makes this distinction concrete. Google Maps grounding achieves high citation density for place-of-interest queries. For marine science evidence queries — the same domain, but requiring scientific rather than place-level grounding — Maps grounding resolves 0 of 25 citations to scientific or dataset sources. NOAA fisheries data, the Center for Whale Research census, Chinook salmon migration corridors, and acoustic hydrophone recordings are all generated as claims but remain entirely uncited. The system knows about these phenomena from training; it cannot bind its claims to verifiable data artifacts.

R_{uncited} — the fraction of scientific sentences in a system’s output with no bound citation — is the measurement primitive that makes this gap quantitative, reproducible, and architecture-diagnostic.

2 R_{uncited} as a formal evaluation metric

2.1 Definition

Let S be the set of sentences in a system’s output. Define $S_{\text{sci}} \subseteq S$ as the sentences containing scientific or quantitative markers. For each sentence $s \in S_{\text{sci}}$, let $\text{cited}(s)$ be the indicator that s names at least one entity appearing in the bound citation set C .

$$R_{\text{uncited}} = \frac{|\{s \in S_{\text{sci}} : \neg \text{cited}(s)\}|}{|S_{\text{sci}}|} \quad (1)$$

$R_{\text{uncited}} = 0$ when every scientific sentence names a bound citation.
 $R_{\text{uncited}} = 1$ when no scientific sentence does. It is undefined when $|S_{\text{sci}}| = 0$.

2.2 Properties

R_{uncited} has four properties that make it useful as an evaluation primitive.

Claim-level granularity. The metric operates at the sentence level. A response with 10 scientific sentences and 9 bound citations still has $R_{\text{uncited}} = 0.1$, distinguishing it from response-level pass/fail metrics.

Architecture-diagnostic. Two systems with the same surface accuracy can have very different R_{uncited} values depending on whether their reasoning traces are passed as context to the language model. The metric discriminates between evidence-bound and parametric-knowledge-based claim generation.

Reproducible. R_{uncited} is computed from the raw API output. No human annotation is required. The scientific marker set is implemented as a regex over domain-specific technical terms.

Model-independent. R_{uncited} can be applied to any language model’s output. It measures the binding between output and context, not internal model behavior.

2.3 Relation to prior evaluation frameworks

The Ragas framework [4] evaluates faithfulness (whether claims are supported by retrieved context), answer relevancy, and context precision. These metrics require a reference context. R_{uncited} measures a weaker but more deployable condition: whether the model’s output names sources in the context it was given, without requiring a predefined reference set.

The ALCE benchmark [5] evaluates citation generation at the claim level. R_{uncited} is complementary: it measures the rate of claims carrying no citation at all, regardless of citation validity.

3 Benchmark methodology and results

3.1 Setup

The benchmark uses the Gemini Interactions API (gemini-3.5-flash, Api-Revision: 2026-05-20) with `google_maps` tool at San Juan Islands coordinates (48.5465, −123.03). Eight scenarios run in parallel via Python `asyncio` (`tools/testing/grounding_parallel_rag.py`), each submitting a distinct query with a distinct RAG context type. Run date: 2026-06-24.

3.2 Scenarios

#	Context injected	Query type
1	Maps-only (none)	Where are SRKWs seen + evidence
2	Live gate data	Same + gate framing
3	Cell provenance	Cell intensity + kernels
4	Surface planner step-log	Gates blocking promotion
5	Maps-only (none)	Sighting check
6	Sighting-assist output	Same + orcast data
7	Maps-only (none)	Shore whale watching plan
8	Hotspot probability data	Same + orcast forecast

3.3 Results

Scenario	Context	R_{uncited}
4	Planner step-log	0%
8	Hotspot data	75%
1	Maps-only orca	60%
7	Maps-only trip	100%
2	Gate context	100%
3, 5, 6	Provenance / sighting	100%

Scenario 4 achieves $R_{\text{uncited}} = 0\%$ by artifact transformation, not citation generation. The query “which gates block promotion” is fully answered from the step-log artifact references without generating science claims about orca ecology. Scenario 8 provides 25 pp lift because hotspot data partially substitutes for science claims. Gate and provenance context (Scenarios 2–3) provide no lift because the injected data does not cover the ecology claims Gemini independently generates.

4 The RAG lift hierarchy as a grounding architecture diagnostic

The 8-scenario results reveal a diagnostic structure:

Context type	R_{uncited}
Step-log (complete reasoning trace)	0%
Structured domain data (hotspots)	75%
Maps-only baseline	60–100%
Narrow structured data (gates)	100%

This is not a function of data volume but of structural alignment between context type and the science claims the model otherwise generates.

Full grounding (step-log): The planner step-log contains the complete record of skills dispatched, panels selected, and artifacts referenced. The query is fully answerable from step-log artifact references without generating science claims. Every output claim refers to an artifact present in the context: R_{uncited} collapses to 0% because $|S_{\text{sci}}| = 0$.

Partial grounding (hotspot data): Structured probability data partially substitutes for the model’s generic science claims. The remaining 75% uncited rate corresponds to ecology claims the hotspot data does not cover.

No grounding (gate data): Gate data covers statistical test outcomes. The model’s response generates broader ecology claims that gate data does not address.

The hierarchy functions as a design diagnostic: context that binds the query to a complete reasoning trace eliminates uncited claims; narrowly structured context does not. This diagnostic is portable to any domain where the interaction’s reasoning trace can be injected as context. The eRAG approach to retrieval evaluation [17] would yield the same diagnostic: only step-log documents are fully utilized by the model in answering the query.

The Citation-Enhanced Generation framework [14] demonstrates that post-hoc citation binding reduces hallucination in chatbots. The hierarchy finding complements this: pre-hoc trace injection eliminates the class of queries that citation binding must address.

5 Extension to AI planning systems

5.1 The step-log as a planning trace

The surface planner step-log that achieves $R_{\text{uncited}} = 0\%$ in Scenario 4 has the structure of a planning trace. The agent received a query, resolved a configuration, dispatched skills, and produced a plan. The step-log records every intermediate state: which skills ran, what their outputs were, and what the final plan consists of. When this trace is injected as RAG context, the language model answers from the trace’s artifact references rather than from world knowledge. $R_{\text{uncited}} = 0\%$ because the trace transforms the query from open-domain science to closed artifact-reference.

The paper “Reasoning with Language Model is Planning with World Model” [6] explicitly frames LLM reasoning steps as world-state traces and proposes evaluating whether those states are accurate. The evaluation problem R_{uncited} addresses is the evidence-binding dimension of this: not whether the intermediate states are accurate, but whether each

state is traceable to the observations that produced it. TRACE [20] demonstrates that constructing knowledge-grounded reasoning chains from structured context eliminates irrelevant retrieval noise in multi-hop queries; the orcast step-log is this structure applied to skill-dispatch interactions. Chain-of-Thought evaluation [21] confirms that intermediate reasoning steps are causally important for correct final outputs, supporting the claim that step-log quality measurement is foundational to planning system evaluation.

5.2 Applying R_{uncited} to planning systems

The benchmark methodology in Section 4 is directly portable to any AI planning system whose interactions produce a structured trace. For a planning system that receives a query about a physical or scientific domain, dispatches skills or sensors to gather current-state information, produces a plan or action sequence based on gathered context, and narrates the plan with references to gathered artifacts: inject the planning trace as RAG context to a language model, ask a domain-specific question, and measure R_{uncited} on the response. When the trace binds the query to artifact references, R_{uncited} approaches 0%. When the trace is sparse or unstructured, R_{uncited} remains high.

The RAG lift hierarchy (Section 5) provides the diagnostic: step-log context achieves full grounding; structured domain data achieves partial lift; narrow structured data achieves no lift. This hierarchy is a property of context-type alignment with the science claims the model would otherwise generate, not a property of context volume. The Ragas framework [4] measures faithfulness separately from answer relevancy; R_{uncited} measures a simpler, deployable condition that precedes both: whether claims are cited at all.

6 Limits and falsification

6.1 Scope limits

Domain specificity. The benchmark uses marine science queries. Extension to other domains requires updating the science marker regex for that domain’s technical vocabulary.

Model dependence. Results were produced with gemini-3.5-flash (2026-06-24). A model with stronger domain-specific knowledge may generate fewer science claims.

R_{uncited} **does not measure semantic accuracy.** A claim that is cited to a bound artifact but semantically wrong is not captured. Extension to factual accuracy requires a reference ground truth.

Step-log quality dependence. Scenario 4 achieves 0% because the surface planner step-log is complete and structured. A sparse or malformed step-log would not achieve the same result.

6.2 Falsification conditions

Claim	Falsifying condition
Step-log injection achieves $R_{\text{uncited}} = 0\%$	Step-log-injected query produces uncovered science claims
Hierarchy is stable across query types	Different Scenario 4 query yields high R_{uncited}
Gate context achieves no lift	Gate data injection covers generated science claims
R_{uncited} diagnoses architecture quality	Two architecturally different systems produce identical hierarchies

6.3 Extension path

Multi-domain validation: run the benchmark on pedestrian routing (pax), clinical care, and manufacturing. World model evaluation: apply the benchmark to JEPA-based world models (V-JEPA [3]) and measure whether planning trace injection reduces R_{uncited} . Semantic accuracy addition: add a second metric for whether cited claims are factually accurate relative to their bound artifacts.

6.4 The benchmark gap in existing physical-world evaluation

Physical-world AI benchmarks evaluate task completion or prediction accuracy without auditing intermediate claim binding. PHYRE [2] measures physical reasoning via puzzle solve rate. STAR [19] evaluates situated reasoning in real-world videos via question answering accuracy. ThreeDWorld [9] measures embodied planning by object transport success rate. TravelPlanner [22] evaluates language agent planning by plan validity. The comprehensive Embodied AI survey [13] categorizes evaluation as navigation, interaction, manipulation, and reasoning accuracy. None measures whether intermediate reasoning claims are bound to their generating sensor observations. EWOK [16] tests world modeling via plausibility matching. R_{uncited} provides the missing primitive.

6.5 World model evaluation as a future extension

A natural extension is to apply this benchmark to world model planning systems whose design explicitly requires sensor-grounded intermediate representations. LeCun’s autonomous machine intelligence architecture [7] specifies that the world model module must learn to represent the information content of the percept that is relevant for predicting future states — a grounding requirement on intermediate representations. V-JEPA [3] and Image World Models [18] are evaluated on downstream accuracy on image and video benchmarks; no claim-level evidence binding metric is proposed. Whether R_{uncited} applied to JEPA planning traces produces the same diagnostic hierarchy as the orcast benchmark — step-log injection achieving near-zero, narrow data injection achieving no lift — is an open empirical question. The benchmark methodology in Section 4 requires only a structured planning trace as input; running it on world model outputs is a direct application with no architectural changes.

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Appendix: Figures

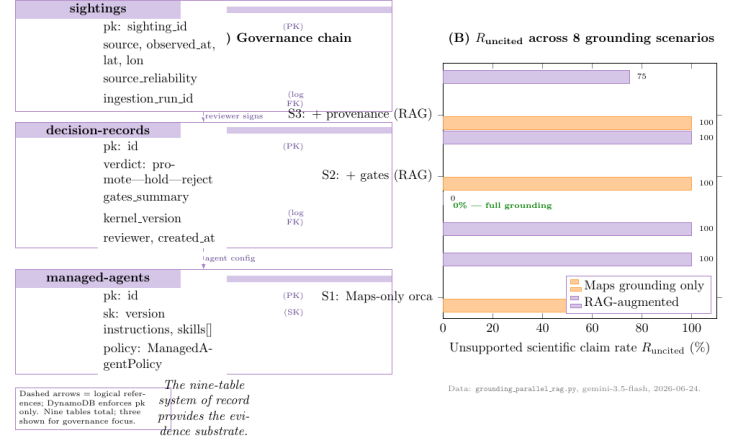


Figure 1: **Evidence-binding failure and its measurement.** (A) The orcast governance chain: sightings, decision-records, and managed-agents in DynamoDB. (B) R_{uncited} across eight grounding scenarios. Maps-only baseline: 60–100%. Surface planner step-log (Scenario 4): $R_{\text{uncited}} = 0\%$. Data: grounding_parallel_rag.py, gemini-3.5-flash, 2026-06-24.

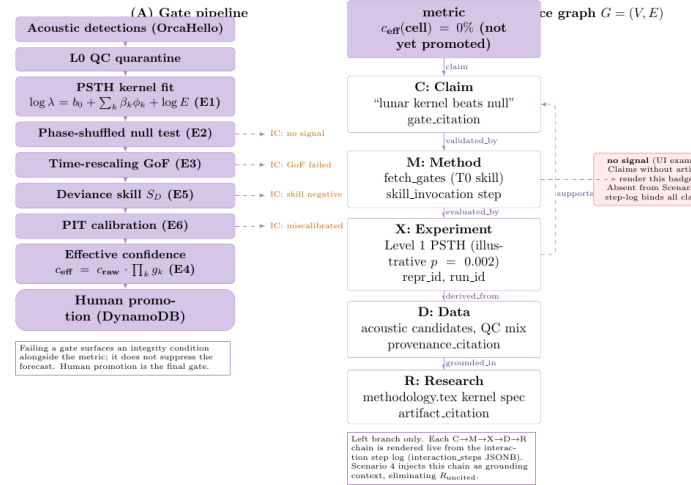


Figure 2: **The mechanism that produces $R_{\text{uncited}} = 0\%$.** (A) Gate pipeline from acoustic detections to displayed confidence. (B) Provenance graph $G = (V, E)$: Claim (C) → Method (M) → Experiment (X) → Data (D) → Research (R). Claims without a bound artifact render the “no signal” badge. Scenario 4 eliminates R_{uncited} because the planner step-log binds every claim to an artifact reference.

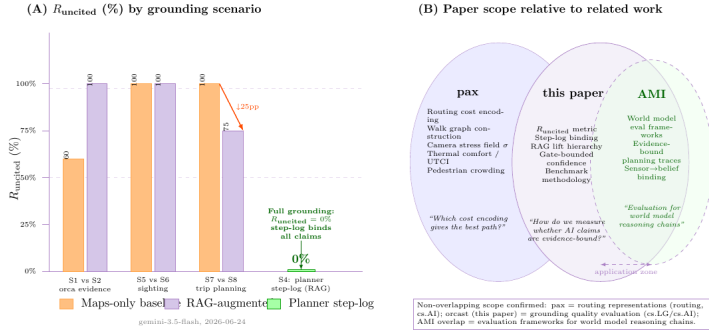


Figure 3: **Grounding quality hierarchy and paper scope.** (A) R_{uncited} for baseline vs RAG-augmented scenarios. The surface planner step-log (green, 0%) dominates; hotspot data provides partial lift (S7→S8); gate context provides no lift. (B) Paper scope: this paper addresses grounding evaluation methodology (center), distinct from pedestrian routing cost representations (pax, left), and directly applicable to world model evaluation infrastructure (AMI, right).